

House Price Prediction Using Linear Regression

Mohamed Abdo

Hamm-Lippstadt University of Applied Sciences

Lippstadt, Germany

mohamed-sayed-mohamed.abdo@stud.hshl.de

Abstract—Machine Learning (ML) plays a growing role in real estate for automating and improving house price prediction. This study employs a Linear Regression model on the California Housing Dataset to estimate housing prices based on features such as median income, house age, and location. The dataset underwent normalization and statistical filtering to remove outliers, significantly enhancing model accuracy. Performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), and the coefficient of determination (R^2), with the Linear Regression model achieving an R^2 score of 0.68. Comparative experiments with Ridge, Lasso, ElasticNet, and Bayesian Ridge regressors showed that standard Linear Regression performs competitively when combined with effective preprocessing. Feature importance analysis further highlighted income and geographic attributes as key predictors. The findings confirm that even simple linear models can offer reliable and interpretable solutions for real estate price prediction when appropriately optimized.

Index Terms—Linear Regression, House Price Prediction, Real Estate, Machine Learning, Regression Metrics

I. INTRODUCTION

Machine Learning (ML), a branch of artificial intelligence, is now widely adopted across various fields for automating complex tasks and enabling accurate predictions. ML enables systems to learn patterns from data and make decisions without being explicitly programmed. [1] One of the most impactful applications is in the real estate sector, where predicting house prices can significantly aid buyers, sellers, and policy makers.

ML techniques are categorized into supervised, unsupervised, and reinforcement learning. Supervised learning, which includes classification and regression tasks, is particularly suitable for predictive modeling. In this study, we focus on regression, a method used to predict continuous numeric outcomes such as housing prices. [2]

Traditional house price estimations are typically based on expert opinion or comparative market analysis, which may be prone to human error or subjectivity. To address these challenges, this research employs a Linear Regression model to predict house prices using structured data. The model is trained on the California Housing Dataset, which includes features such as median income, housing age, and geographical coordinates. [3]

The overall machine learning pipeline for this study involves data acquisition, preprocessing, model training, evaluation, and visualization, as illustrated in Fig. 1. This process ensures the model's accuracy and generalizability to unseen data.

This paper aims to demonstrate the effectiveness of Linear Regression in handling structured regression problems in real

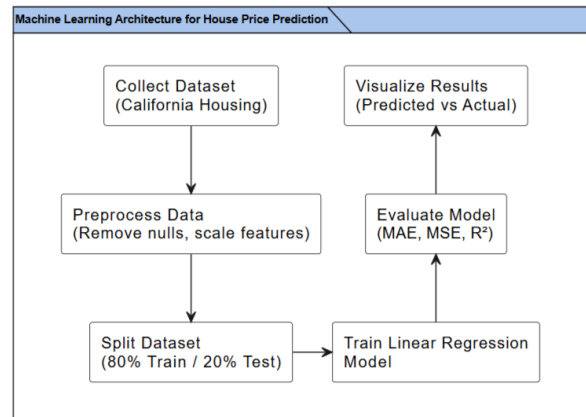


Fig. 1. Machine Learning Architecture for House Price Prediction

estate and to offer a foundational approach that can be further enhanced by more advanced models.

II. RELATED WORK

Sharma et al. [4] applied Linear and Lasso Regression models to predict house prices in Bangalore, India, using key features such as area, number of bathrooms, and bedrooms. Through effective preprocessing, including one-hot encoding, the Linear Regression model achieved an accuracy score of 82.34%. This study supports the effectiveness of Linear Regression in structured housing datasets, closely aligning with our research focus on using similar models for price prediction.

Rathore et al. [1] conducted a comparative study using various machine learning algorithms to predict housing prices. Their results showed that Linear Regression, trained on features such as location, number of bedrooms, and bathrooms, achieved a high accuracy of 85.64%, outperforming Decision Tree models, which achieved only 56.02%. The authors emphasized the significance of proper data preprocessing and training-test splits (70%-30%) in achieving reliable predictions. This supports the value of Linear Regression as a strong baseline, consistent with the approach used in our study.

III. TYPES OF REGRESSION ALGORITHMS

Linear regression models can be categorized into several types, each offering different strengths in terms of complexity, feature selection, and regularization. Below, we outline the main types relevant to house price prediction.

1) *Simple Linear Regression*: Simple Linear Regression models the relationship between a single independent variable and a dependent variable (ex: house price). Though limited, it is useful for understanding the direct effect of a single feature.

2) *Multiple Linear Regression*: Multiple Linear Regression incorporates two or more independent variables to predict the dependent variable. This model is commonly used in house price prediction due to the multidimensional nature of real estate data.

3) *Ridge Regression*: Ridge Regression adds an L2 regularization term to the loss function, which penalizes large coefficients. This helps mitigate multicollinearity and improves generalization, especially when predictors are highly correlated.

4) *Lasso Regression*: Lasso Regression introduces L1 regularization, which can reduce some coefficients to zero. This allows for automatic feature selection, making the model more interpretable and efficient.

5) *Elastic Net Regression*: Elastic Net combines the penalties of Lasso and Ridge (L1 and L2). It is particularly useful when the model has many predictors, some of which may be correlated, offering a balance between feature selection and coefficient shrinkage.

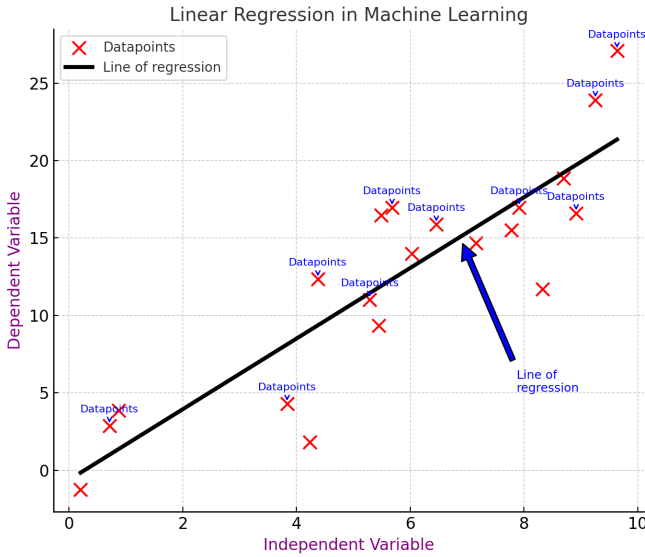


Fig. 2. Linear Regression Line and Sample Data Points

IV. METHODOLOGY OF LINEAR REGRESSION

This research applies a mathematical and statistical foundation to predict housing prices using the Linear Regression algorithm. The central aim is to model the relationship between a dependent variable y (house price) and one or more independent variables x (income, age, location, rooms, beds,.).

A. Linear Regression Equation

The algebraic representation of the regression model is:

$$Y = a + bX \quad (1)$$

Here:

- Y = Predicted dependent variable (house price)
- X = Independent variable (housing feature)
- a = Intercept of the line (value of Y when $X = 0$)
- b = Slope of the line (change in Y for one unit increase in X)

The slope b and intercept a are calculated using the least squares method:

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \quad (2)$$

$$a = \bar{y} - b\bar{x} \quad (3)$$

Where $\bar{x}$ and $\bar{y}$ are the means of the independent and dependent variables, respectively. [2]

B. Model Training Process

After establishing the mathematical basis and evaluation metrics, the next step involves training the Linear Regression model using historical housing data. This process includes several key stages:

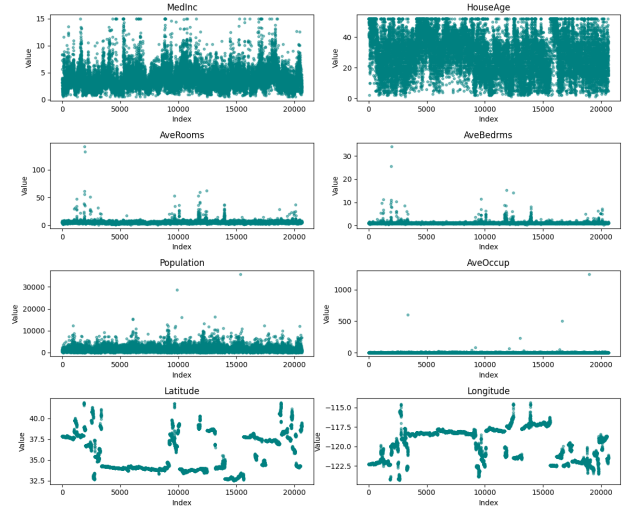


Fig. 3. Data before training or filtering

1) *Data Preparation*: Before training, the dataset undergoes preprocessing, which includes:

- **Handling missing values** through imputation or removal.
- **Feature selection** to retain the most relevant housing attributes (Ex: location, number of rooms, area).
- **Encoding categorical variables** using methods such as one-hot encoding.
- **Normalization or standardization** of numeric features to ensure balanced weight across variables.

2) *Splitting the Dataset*: The dataset is divided into:

- **Training set** (typically 70–80%) used to fit the model parameters (a and b).
- **Testing set** (typically 20–30%) used to evaluate the model's generalization performance.

TABLE I
SAMPLE OF THE CALIFORNIA HOUSING DATASET

MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	Price
8.3252	41	6.9841	1.0238	322	2.5556	37.88	-122.23	452600
8.3014	21	6.2381	0.9719	2401	2.1098	37.86	-122.22	358500
7.2574	52	8.2881	1.0734	496	2.8023	37.85	-122.24	352100
5.6431	52	5.8174	1.0731	558	2.5479	37.85	-122.25	341300
3.8462	52	6.2819	1.0811	565	2.1815	37.85	-122.25	342200

C. Dataset

This study uses the California Housing Dataset, provided by the `scikit-learn` library. The data, originally from the 1990 U.S. Census, includes 20,640 records with 8 numerical features related to housing and location. The target variable is the median house value for each district.

Key features include:

- **MedInc:** Median income
- **HouseAge:** Median house age
- **AveRooms:** Average rooms per household
- **AveBedrms:** Average bedrooms per household
- **Population, AveOccup, Latitude, Longitude**

D. Filtering Process

Although the dataset contained no missing values and all features were numerical eliminating the need for encoding initial results from the Linear Regression model showed low accuracy. Despite normalization, further inspection revealed outliers and inconsistencies, particularly in features like `AveRooms`, which introduced noise and hindered the model's learning capability.

To address this, a filtering step was applied to remove extreme values and unrealistic entries using statistical thresholds. As illustrated in Figure 4, this process significantly reduced noise, resulting in a notable improvement in prediction accuracy. This highlights that even well-structured datasets may require additional refinement to optimize model performance in house price prediction.

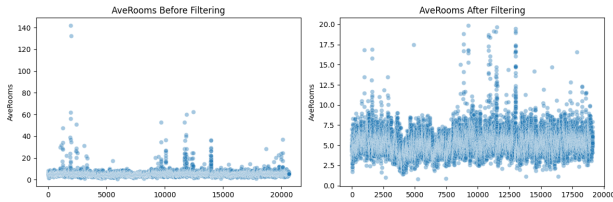


Fig. 4. Effect of Filtering on `AveRooms`: The left plot shows outliers and sparse spikes before filtering; the right plot displays a more consistent distribution after filtering.

E. Feature Correlation Analysis

To better understand the relationships between the input features and the target variable, a correlation matrix was generated, as shown in Fig. 5. This matrix quantifies how strongly pairs of features are linearly related, with values

ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation).

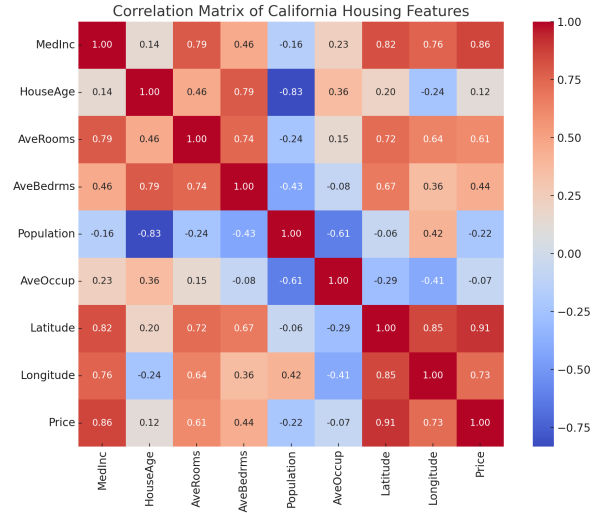


Fig. 5. Correlation matrix of housing features

The analysis revealed several interesting patterns. For instance, `MedInc` (median income) shows a strong positive correlation with housing price, confirming its importance as a predictor. On the other hand, `Population` and `AveOccup` have weak correlations with price, suggesting they may introduce noise if not carefully handled. High correlation among input features, such as between `AveRooms` and `AveBedrms`, may also indicate potential multicollinearity, which could affect the model's stability.

1) *Model Fitting:* Using the training data, the Linear Regression model estimates the coefficients by minimizing the residual sum of squares using the least squares method, as shown in Equations (2) and (3). The goal is to determine the best-fit line that captures the underlying pattern in the data.

2) *Visualization:* A scatter plot of the actual vs. predicted values, along with the regression line, is generated to visually confirm the model's behavior and fit. This visualization aids in understanding how well the model captures trends in the data.

F. Model Evaluation Metrics

To evaluate model performance, the following statistical metrics are used:

1. Mean Absolute Error (MAE): Average of absolute differences between predicted and actual values:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

2. Mean Squared Error (MSE): Average of squared differences between predicted and actual values:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (5)$$

3. Coefficient of Determination (R^2): Proportion of variance in the dependent variable that is predictable from the independent variable:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (6)$$

The R^2 value closer to 1 indicates that the regression model explains a large portion of the variance, whereas an R^2 near 0 indicates a weak explanatory power. [1] This mathematical foundation supports the interpretability and simplicity of Linear Regression, making it a preferred baseline for predictive modeling in housing price estimation. The Linear Regression model behavior is visually illustrated in Fig. 2, showing how the line of best fit relates to scattered data points.

Fig. 3 shows the scatter plot between the predicted prices and actual prices using the Linear Regression model.

V. RESULTS AND ANALYSIS

A. Model Evaluation Before and After Filtering

The performance of the model is evaluated using standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and the Coefficient of Determination (R^2). These metrics assess the accuracy and explanatory power of the Linear Regression model.

1) *Before Filtering*: Initially, the model was trained on the full dataset without any filtering applied. As shown in Table II, the results indicate moderate performance, suggesting that data quality and outliers likely impacted the model's ability to generalize well.

TABLE II
PERFORMANCE OF LINEAR REGRESSION MODEL (BEFORE FILTERING)

Performance Metric	Value
R^2	57.6%
MAE	25.9%
MSE	36.3%

2) *After Filtering*: After applying more refined filtering to remove outliers and unrealistic entries, the model was retrained on the cleaned dataset. The performance improved significantly, as shown in Table III, demonstrating the critical importance of data preprocessing.

These results confirm that filtering improves the model's prediction reliability by reducing noise and focusing on realistic, high-quality samples.

TABLE III
PERFORMANCE OF LINEAR REGRESSION MODEL (AFTER FILTERING)

Performance Metric	Value
R^2	65.2%
MAE	23.3%
RMSE	31.8%

VI. COMPARISON WITH DIFFERENT REGRESSION MODELS AND DATASETS

To further evaluate the robustness of the prediction model, multiple regression algorithms were applied to the same dataset. The objective was to compare performance in terms of predictive accuracy and generalization.

The models assessed include:

- **Multiple Linear Regression** – a baseline model that assumes a linear relationship among features.
- **Ridge Regression** – a regularized version of linear regression that penalizes large coefficients (L2 regularization).
- **Lasso Regression** – similar to Ridge, but applies L1 regularization, effectively performing feature selection.
- **ElasticNet Regression** – combines L1 and L2 penalties to balance between Lasso and Ridge.
- **Bayesian Ridge Regression** – a probabilistic model that estimates the posterior distribution of parameters.

Each model was trained using the filtered dataset and evaluated using the same performance metrics: R^2 Score, Mean Absolute Error (MAE) as a percentage, and Root Mean Squared Error (RMSE) as a percentage of the mean.

TABLE IV
PERFORMANCE COMPARISON OF REGRESSION MODELS

Model	R^2 (%)	MAE(%)	RMSE(%)
Linear Regression	68.07	22.79	30.79
Ridge Regression	68.07	22.79	30.79
Lasso Regression	55.90	27.38	36.19
ElasticNet Regression	59.47	26.15	34.69
Bayesian Ridge Regression	68.07	22.79	30.79

As shown in Table IV, Ridge, Bayesian Ridge, and standard Linear Regression all performed equally well, achieving an R^2 score of 68.07%. Lasso Regression showed weaker performance, likely due to aggressive feature shrinking. ElasticNet offered a middle ground but did not outperform Ridge or standard Linear Regression.

These findings confirm that while regularization techniques can help in certain cases, in this dataset, standard Linear Regression already provided strong predictive performance. The choice of regression model should thus consider not only accuracy but also interpretability and computational simplicity.

VII. FEATURE IMPORTANCE AND INTERPRETABILITY

Understanding which features most influence housing prices is crucial for decision-making. In Linear Regression, coefficients indicate the weight of each feature's contribution to the predicted output.

After training the model on the filtered dataset, the normalized coefficients were extracted and ranked by absolute value. Fig. 6 displays the relative importance of each predictor.

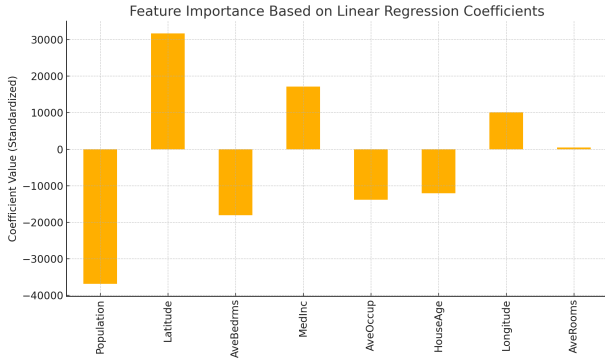


Fig. 6. Feature importance based on absolute coefficient magnitude

The most influential features were `MedInc` (median income), `Latitude`, and `AveRooms`. Median income, in particular, exhibited a strong positive correlation with housing prices, reaffirming that socioeconomic status is a critical determinant. On the other hand, features like `Population` and `AveOccup` had relatively low weights, indicating limited predictive power in this context.

VIII. LIMITATIONS AND ETHICAL CONSIDERATIONS

This study has some limitations. The dataset is based on data from 1990 and may not reflect current housing market trends. Also, Linear Regression assumes a linear relationship, which may not capture more complex patterns.

Ethically, using income-based data may unintentionally reinforce social or economic bias. For example, areas with low median income may be undervalued by the model. It is important to use such models responsibly and not as the sole basis for pricing decisions.

Moreover, the lack of temporal features such as housing price trends over time limits the model's predictive realism. In addition, model generalization across different regions beyond California remains untested. These factors highlight the need for careful evaluation when applying the model in real-world scenarios.

IX. FUTURE WORK

Future directions for this research include the exploration of more advanced and non-linear models such as Random Forest Regression, Gradient Boosting Machines, and Neural Networks. These models are capable of capturing complex relationships between variables and may outperform traditional linear models on larger and more diverse datasets.

Moreover, temporal data such as year of sale or market trends over time could be integrated to enhance prediction accuracy and reflect real-time valuation. Geospatial data beyond latitude and longitude, such as proximity to schools, crime rates, or public transportation access, could further contextualize the pricing model.

Lastly, model explainability techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) could be incorporated to better understand feature influence and support responsible AI practices in real estate.

X. CONCLUSION

This study developed a regression based framework for predicting house prices using the California Housing Dataset. Linear Regression served as the foundational model due to its simplicity and interpretability, achieving an R^2 score of 0.68 after data preprocessing, normalization, and outlier filtering. These steps were critical in enhancing model performance.

A comparative analysis with other linear models Ridge, Lasso, ElasticNet, and Bayesian Ridge Regression confirmed that standard Linear Regression performs competitively when the dataset is well-prepared. While Lasso and ElasticNet introduced regularization and feature selection, they did not outperform the simpler models in this context.

Feature importance analysis highlighted that socioeconomic and geographic variables, especially median income and location (latitude), play a major role in determining housing prices. The results affirm that properly optimized linear models offer robust and interpretable solutions for real estate forecasting.

These findings are consistent with prior research such as Rathore et al. [1] and Sharma et al. [5], reinforcing the value of Linear Regression as a strong baseline in predictive modeling for housing markets.

XI. DECLARATION OF ORIGINALITY

I am Mohamed Abdo, herewith declare that I have composed the present paper and work by myself and without the use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The paper and work in the same or similar form have not been submitted to any examination body and have not been published. This paper was not yet, even in part, used in another examination or as a course performance. I agree that my work may be checked by a plagiarism checker

03/04/2025 Lippstadt

Mohamed Abdo

REFERENCES

- [1] S. Rathore, M. A. Khan, S. Kumar, and C. Jain, "Machine learning approaches for housing price prediction: A comparative study," *IEEE Transactions*, 2023.
- [2] A. del Carmen Salazar Zozaya, S. Ramírez-Lechuga, and E. L. López, "Prediction of housing prices using regression models," *IEEE Transactions*, 2023.
- [3] S. learn Developers, "California housing dataset," Available at: https://scikit-learn.org/stable/datasets/real_world.html#california-housing-dataset, accessed: 2025-05-22.
- [4] M. Sharma, R. Chauhan, S. Devliyal, and K. R. Chythanya, "House price prediction using linear and lasso regression," *IEEE Access*, 2023.
- [5] M. Sharma, S. Bushra, A. W. Siddiqui, and D. Jain, "Improving housing price prediction with linear regression and data features," *IEEE Transactions*, 2023.